

Dilemmadata: On the Inter-operability of Heterogeneous Roman Numeral Datasets

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Introduction

The boom of deep learning and big data approaches in recent years has heavily impacted many fields, including digital musicology. Recent efforts have brought forth large-scale datasets of harmony-annotated scores enabling longitudinal studies and machine-learning approaches, notably the AugmentedNet dataset (AND) (Nápoles López et al., 2021) and the Distant Listening Corpus (DLC) (Hentschel et al., 2025). Although both encode human expert analyses in the form of Roman numeral labels, in practice however, harmonising them into a single training and research corpus proved cumbersome and error-prone due to disparate encoding paradigms. We present the outcome of this effort: `dilemmadata`, the largest homogeneous Roman numeral dataset available to date. The name reflects the difficult yet unavoidable decisions necessitated by the reconciliation of such carefully crafted label sets. In Section [1](#), we will briefly introduce the source datasets and their underlying encoding paradigms, before discussing our solution to this reconciliation challenge (Section [2](#)) and the learnings for the music encoding community (Section [3](#)).

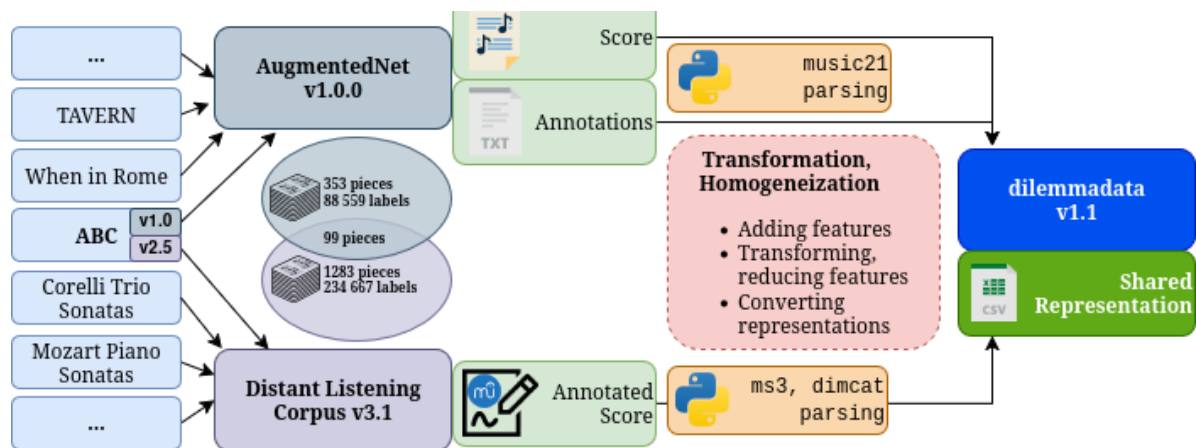


Figure 1. Data processing Pipeline from left to right. The left side depicts the AugmentedNet dataset and the Distant Listening Corpus as meta-corpora comprising multiple sub-corpora which comprise multiple sub-corpora in their own right, one of them identical: the Annotated Beethoven Corpus, ABC, accounting for 70 of the 99 overlapping pieces (remaining overlaps stem from distinct sources and provide alternative, not identical analyses). Each dataset’s music representations (green rectangles) are processed with their respective parsing library (orange rectangles) while applying multiple transformations (red rectangle). The pipeline yields our aligned and homogenised dataset.

1 Datasets

1.1 The AugmentedNet Dataset (AND)

AugmentedNet¹ is an automatic Roman numeral analysis neural network developed by Néstor Nápoles López as part of his PhD research (Nápoles López, 2022). The accompanying dataset comprises 353 distinct pieces of tonal music, each annotated at the segment level in the RomanText (‘.rntxt’) format—a human- and machine-readable plain-text standard that encodes chord root, quality, inversion, added intervals, and analyst notes, with full support for bidirectional conversion in `music21` (Cuthbert & Ariza, 2010; Tymoczko et al., 2019). Source scores originate from diverse formats (MusicXML, Humdrum, ABC), and export as TSV slice-segmented tables, including fields for onset, duration, beat position, and Roman numeral labels. Dataset construction aggregates analyses from six publicly available corpora—ABC, BPS, HaydnSun, TAVERN, WiR and WTC (Chen & Su, 2018; Devaney et al., 2015; Gotham et al., 2023; Nápoles López, 2017). Many works on automated Roman numeral analysis also train and compare to the AND, effectively establishing it as a standard baseline for Roman numeral prediction models (Karystinaios & Widmer, 2023; Sailor, 2024).

1.2 The Distant Listening Corpus (DLC)

The DLC² provides 1,283 pieces from four centuries, symbolically encoded in MuseScore’s MSCX format and enriched with embedded DCML harmony

¹ <https://github.com/napulen/AugmentedNet>

² https://github.com/DCMLab/distant_listening_corpus

annotations, including Roman numerals, figured bass symbols, and chord labels, directly integrated as text and symbol objects within the score files (Hentschel et al., 2025). These annotated scores have been collaboratively created and reviewed by a team of trained music theorists using a distributed curation pipeline on GitHub (Hentschel, Moss, et al., 2021), ensuring notational well-formedness through automated `ms3` validations and inter-annotator consensus. Each sub-corpus, spanning more than 40 thematic collections from J.S. Bach’s English Suites to N. Medtner’s Tales, is released as both a GitHub repository and a frictionless data package (Fowler et al., 2018), with TSV exports of notes, chords, and harmonies available for downstream computational use. We removed 15 pieces overlapping with AND (see Section 2.3), leaving 1,268 annotated DLC pieces.

Figure 2 displays measures 10-13 from the first movement of L. v. Beethoven’s Piano Sonata No. 3 in C major, Op. 2, No. 3. The musical score is shown in 4/4 time, with dynamics *sf* and *ff*. Below the score are three layers of harmonic annotations:

- AugmentedNet Dataset:** I, V⁶₅, I, vii^o7/V, I⁶₄, V⁷, I
- Distant Listening Corpus:** I⁶₄, I, V⁶₅, V⁶₅, I⁶₄, I, vi⁶, vii^o/VV(^{#0}₄), V(⁹₄), V⁷, I|PAC}
- dilemmadata (simple):** I, V⁷_V, V⁷, I, vii^o₇⁶, vii^o, Cad, V⁷, I

Figure 2. Measures 10-13 from the first movement of L. v. Beethoven’s, Piano Sonata No. 3 in C major, Op. 2, No. 3. The three annotation layers show labels and harmonic rhythm from the two source datasets and, in the bottom system, of our respective simplified representation without inversions and secondary keys (which are represented elsewhere in our dataset).

2 Dataset Alignment

2.1 Disparate encoding paradigms

The AND relies on the RomanText encoding standard and paradigm where Roman numeral analyses are stored as stand-off text files that can be paired with digital score encodings using `music21` (Cuthbert & Ariza, 2010; Tymoczko et al., 2019). DLC annotations, on the other hand, are encoded directly within MuseScore files, adhere to the DCML harmony annotation

standard, and can be parsed and analysed using `ms3` and `dimcat` libraries (Hentschel et al., 2023; Hentschel, Neuwirth, et al., 2021; Hentschel & Rohrmeier, 2023; Neuwirth et al., 2018). As shown in Figure 1, we employed the respective Python libraries to transform the datasets into a shared CSV representation containing one row per note, including annotation label. This involved adapting the original AugmentedNet processing pipeline, bypassing its creation of fixed-size, sixteenth-note slicing windows in favour of a note-wise representation (thereby expanding the dataset from roughly 100K annotated slices to over 750K individual annotated notes). For the DLC, we achieved the same representation by unfolding repeats and joining the tabular note and annotation data.

2.2 Disparate annotation standards

Figure 2 illustrates the primary discrepancies between the RomanText and the DCML harmony annotation standards. Between the two analyses, the one from the DLC has a perceptibly higher granularity and, thereby, a faster harmonic rhythm, attributable partly to the DCML standard’s elaborate syntax for suspensions (expressed within parentheses). Consequently, the two standards yield analyses that frequently diverge in terms of chord types and which notes are to be considered as chord vs. non-chord tones. Whereas in RomanText, cadential 6/4 chords are annotated as either **I64** or **Cad64**, the DCML standard consistently encodes them as **V(64)** suspension chords, differentiating them from **I64** inverted tonic harmonies. The chord type syntax is slightly different (e.g. $\emptyset$ vs. $\%$ for the half-diminished seventh chord) and, complicating matters further, the AND assembles datasets that differ in their usage of the RomanText syntax.

We addressed these discrepancies by distributing the encoded information of each annotation label across dedicated feature columns such as root, bass, chord type, local key, etc. This approach enables users to flexibly choose between those features that are covered by both standards and those made available by the DCML standard exclusively (such as suspensions, pedal points, cadences, to name a few). This involved crafting shared vocabularies for each shared feature, followed by careful validation. For example, `dilemmadata` uses a harmonized chord type vocabulary which is the outcome of one-to-one mappings where possible, and manual adjustments (verified against the scores) where necessary. Importantly, we carefully replaced each occurrence of the cadential 6/4 chord, the seventh most frequent harmony, with the symbol **Cad**. Based on the harmonized features, we re-synthesized simplified chord labels (see Figure 2), offering a more general chord representation with a more homogeneous harmonic rhythm (when removing immediate duplicates).

2.3 Overlapping pieces

To reconcile AND and DLC, we compiled per-piece metadata for both collections, documenting available annotation features, provenance, and encoding formats. We then identified overlapping pieces by matching file and

folder names, metadata fields, and, where necessary, through manual inspection of score content. This process yielded 99 pieces common to both datasets, of which 15 were removed because they appeared in the AND test set; the remaining 84 overlaps were retained as shared reference material. To support evaluation and downstream tasks, we also constructed a 20% test set of the DLC by selecting pieces on a per-sub-corpus basis, excluding collections too small for reliable sampling or too skewed towards a single mode.

3 Outcomes and Future Work

The final `dilemmadata` corpus comprises 1,621 pieces (353 from the AND plus 1,268 from the DLC) and over 2.8 million note-wise Roman numeral annotations, each enriched with precise onset, duration, beat-level features, chordal root, quality, inversion, extensions, and contextual metadata. Annotations carry provenance hashes and validity flags to facilitate selective filtering. All TSV files, metadata descriptors, and precomputed test splits are available for download³. For a full list of features, we refer our readers to the README.

Beyond the dataset itself, our contribution demonstrates that discrepancies between Roman numeral datasets produced under disparate paradigms extend far beyond mere syntax; they create severe impediments to integration and interoperability. The substantial effort required to carefully transform annotations for equivalence without distorting the original semantic intent underscores, in our view, the urgent need for a generalised data model capable of interfacing with the diverse harmony encoding standards currently in use (e.g., Temperley & Clercq, 2013; Cambouropoulos, 2016; White & Quinn, 2016; Chen & Su, 2018; Huron, 2020; Hentschel et al., 2022). Ultimately, we propose the overlapping `dilemma` pieces, annotated under both source paradigms, as a catalyst for replacing the prevailing reliance on objective “ground truth” in training scenarios, inviting the encoding community to discuss paths toward contingent, probability-based target labels that better reflect the inherently ambiguous nature of harmonic analysis.

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Statement on the use of generative AI (if applicable)

We have employed a Large Language Model for re-formulating sentences from our original draft in a more concise and idiomatic way.

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³ <https://anonymous.com>

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Dump

Despite their scale, these corpora employ divergent annotation schemas: AugmentedNet uses a combination of multiple digital score encoding formats with stand-off annotation files created as or converted to the RomanText standard, whereas

In response to these issues, we introduce *dilemmadata*, a unified Roman numeral corpus that aligns AugmentedNet and DLC through a joint mapping of their annotation schemas. Our contributions are threefold: (1) the formalization of representational divergences and the design of an algorithmic conversion pipeline preserving musical semantics; (2) a preliminary evaluation of mapping validity; and (3) the release of the largest

merged Roman numeral corpus to date, supporting interoperability, training of deep learning algorithms for automated Roman numeral analysis, and informing future standards development.

2.1 Comparison of the Roman Numeral Standards

Many encodings of tonal harmony have been proposed and used, some of which were conceived to accommodate traditional Roman numeral analysis in one way or another (Cambouropoulos, 2016; Chen & Su, 2018; Huron, 2020; Temperley & Clercq, 2013; Tymoczko et al., 2019; White & Quinn, 2016). These approaches vary in the number of encoded features and their representation. Both factors have a strong influence on an encoding’s interoperability and its practicability in human-expert tasks. For example, a standard that expresses roots as scale degrees and chord tones as enharmonic interval classes such as (Cambouropoulos, 2016) cancels out important information about the chord type (e.g., the difference between a dominant-seventh chord and a German sixth chord). What these approaches gain in generalizability and stylistic independence, they lose in descriptive precision and compatibility with more finegrained standards, which are, in return, downward-compatible.

The most precise and feature-rich standards for encoding ordinary Roman numeral analysis that are currently available are the RomanText format and the DCML annotation standard (first introduced in Neuwirth et al., 2018 and extended in Hentschel, Neuwirth, et al., 2021), the latter being used in the DLC. Both standards are highly similar in the encoded features and their representation, and it is in principle possible to convert between them. Also, both account for the need to come up with parsable chord labels that contain a multitude of features while remaining straightforward for musicologists to read and write. The differences between the two standards mainly concern the respective approaches toward score annotation, which we will explain in the following, showing how they relate to each other and how to increase interoperability between the respective annotations.

Both RomanText and DCML support Roman numeral annotations at arbitrary temporal positions, but differ in their entry and export workflows. RomanText annotations are created as plain-text .rntxt or .txt files—manually or via scripts—and can be aligned with symbolic scores through `music21`. This makes it possible to annotate music even without a pre-existing encoded score and to manage multiple analysts’ versions independently, though it requires explicit specification of temporal positions and context-switching between editor and score. DCML annotations, by contrast, are attributed directly to events in MuseScore files, ensuring accurate timestamps and providing a familiar interface to musicologists/annotators.

The two standards follow different approaches to non-chord tones. DCML explicitly encodes suspensions, neighboring motions, and added tones, distinguishing structural elaborations from superficial ornaments, whereas RomanText represents non-chord tones implicitly by omission from the chord’s pitch-class set. For instance, a cadential six-four appears as V(64)

under DCML, emphasizing its dominant function and the contrapuntal phenomenon of a double chord-tone suspension (that of the fifth and the third), but as I64 (or Cad64) in RomanText, interpreting it as a second-inversion I-chord (albeit without conveying any functional interpretation as a tonic, which would be questionable).

2.2 Reconciliation procedure

Next, we conducted a detailed comparison of each dataset’s annotation schema. For shared label types (in particular, Roman numeral harmony), we catalogued which features were directly compatible, which ones required algorithmic transformation, and which were missing from both corpora. Where transformations risked loss or distortion of musical semantics, such as inversion notation at phrase boundaries or rare chord extensions, we employed validity masks to exclude labels that could not be reliably converted. Based on this analysis, we defined a target schema capturing root, quality, inversion, extensions, and special functions, and implemented a conversion pipeline to map each dataset into this common representation.

Finally, with both corpora mapped to the unified schema, we merged them into a single TSV note-wise table and removed remaining duplicates. To support evaluation and downstream tasks, we then constructed a DLC-based test split comprising roughly 20% of the corpus. Test-set pieces were selected on a per-sub-corpus basis, excluding collections too small for reliable sampling, and only those with complete annotation feature sets were included. The result is a harmonized, interoperable dataset ready for comparative harmonization modeling and further analysis.

2.3 Comparison of the Roman Numeral Standards

Next, we augmented the note-wise table with detailed musical context: time signatures and key signatures for each event; several beat-level features such as ‘mn_onset’ (measure-relative quarter-beat index), ‘beat_float’ (integer on-beats versus fractional off-beats), a cumulative ‘beat_continuous’ count, a boolean ‘is_downbeat’ flag, and a categorical ‘downbeat’ label (1–4); and the ‘note_degree’ in the current key. With this enriched representation in place, we defined our target label set, ‘a_simpleNumeral’, a very reduced label representing chordal root and sonority only, in conjunction with separate fields conveying the chord features of chord inversion, extensions, and applied degrees. Mapping original the AND labels to this schema involved context-sensitive lookups for outliers that failed our regular expression (Listing 1), e.g. “ø” $\mapsto$ “%”. The finalized mapping yielded 132 distinct label types across 88,559 tokens (522 marked as ‘none’), and we applied a targeted post-processing rule: all second inversion tonics immediately preceding a dominant scale degree were recast as Cad chord to even out one of the major discrepancies (cf. 2).

3 Outcomes

while tracking column schema, data types, and human-readable descriptions for each new column. These include precise quarter-note offsets and durations represented as exact fractions and division counts (instead of floats), staff and voice identifiers, and the Git hash of the last commit touching each source file, allowing us to detect upstream changes in future releases.

We processed version 3.1 of the Distant Listening Corpus using the `ms3` Python library to iterate through every piece and sub-corpus, extracting unfolded note events (that is, with expanded repeats) and their embedded annotations directly from the uncompressed MuseScore files (`.mscx`). Each score was parsed into a DataFrame compatible with the information produced with `music21`, to which we added rich metadata columns—division counts (`'divs'`), a boolean `'is_harmony_onset'` flag, and the Git commit hash for provenance tracking. We translated primary and secondary Roman numeral roots into scale-degree representations, normalized chord-type vocabularies and inversion labels (0–3) to match `music21` conventions, and converted scale-relative pitches into absolute pitch values for both local and tonicization contexts. Phrase boundaries were recoded as a boolean column, and we supplemented every note event with comprehensive beat-level features: measure-relative quarterbeat index (`'mn_onset'`), integer versus fractional beat positions (`'beat_float'`), a continuous beat count, a boolean `'is_downbeat'` marker, and a categorical `'downbeat'` label. Finally, each note carries its scale-degree within the current key (`'note_degree'`).

Building on this enriched table, we applied our `'a_simpleNumeral'` mapping to distill Roman numeral annotations to a strippeddown form without inversions, extensions, or secondary degree. Outlier labels that did not conform to our validation regex were contextually translated, e.g., `"b" ↦ " - "`, producing 185 distinct label types across 318,781 tokens (101 labeled as `'none'`). To ensure functional consistency with the AND, we then filtered all V(64) chords and mapped them to the label `Cad`. This pipeline translates the DLC to our unified schema and readies the corpus for interoperable analysis alongside the AND.

instigate a wider discussion on the impact that

Quantifying the impact of these transformations remains challenging. Harmony analysis lacks an objective “ground truth,” and no standard metrics exist for assessing how well a Roman numeral label corresponds to its musical context under different theoretical assumptions. Moreover, the AND itself

aggregates multiple public corpora, combining manual and automated annotations, so annotation variation and smoothing was already present in the dataset. By retaining provenance metadata and flagging labels with potentially reduced fidelity, we enable researchers to identify and, if desired, exclude annotations where semantic drift may have occurred.

Looking ahead, the need to establish context-sensitive validity metrics is a priority. Such metrics should be grounded in theory and able to quantify how appropriate an annotation is depending on the surrounding context. From the perspective of automated Roman numeral analysis systems that create annotations, multi-label prediction could propose alternative analyses and systematic comparisons of competing/correcting Roman numeral annotations.

To support future research needs, we also plan to release a lightweight conversion library that generates tailored joint representations from heterogeneous Roman numeral corpora. Users will be able to choose annotation granularity (time-based or notebased), and a number of annotation output options, such as root-only schemas, full intervallic encodings, and resolve local-key or applied-key features as needed. By lowering the barrier to dataset integration, we aim to create a tool that embraces interoperability and drives the development of shared standards in computational musicology and automated Roman numeral analysis.